RAG: Baseline Retrieval vs. Cross-Encoder Re-Ranking

Ashish | Student ID: 2026909001 | 19 October 2026

Background and objective

Retrieval-augmented generation (RAG) supplies a language model with passages retrieved from a document before it answers. The answer therefore depends in part on which evidence reaches the prompt. This experiment compares direct selection from dense retrieval with a second-stage re-ranker. Its main question is whether re-ranking can bring a relevant clause into the selected three when that clause is already among the initial ten candidates.

Data and method

The corpus is NIST SP 800-63B-4, *Digital Identity Guidelines: Authentication and Authenticator Management* (July 2025), downloaded from NIST's public publication server. The notebook extracts the PDF text page by page and reports 129 pages, 42,918 words, and 457 chunks. Chunks are generated from the extracted text in page-preserving windows of 140 words with 30 words of overlap. The notebook's queries refer to clauses checked in the document.

The bi-encoder, sentence-transformers/all-MiniLM-L6-v2, embeds the query and each chunk separately. Cosine similarity gives the initial top 10. In the baseline, the top three are sent directly to the generator. In the re-ranked pipeline, the cross-encoder, crossencoder/ms-marco-MiniLM-L-6-v2, scores those same ten query–chunk pairs; its three highest-scoring candidates are then used. Both paths use google/flan-t5-base with the same prompt and decoding settings. A bi-encoder is suited to the initial search because chunk vectors can be computed in advance; the cross-encoder jointly scores a query and candidate, at extra per-query cost.

Baseline RAG	Re-ranked RAG
Bi-encoder retrieves 10 chunks; select ranks 1–3 and generate an answer from them.	Use the same top 10; cross-encoder reorders them; select ranks 1–3 and generate an answer from them.

The assessment checks whether a page-verified evidence chunk enters the selected top three. This measures retrieval selection, not answer completeness. Cosine and cross-encoder scores come from different models and should not be compared numerically with each other. Scoring all 457 chunks with the cross-encoder would require query-specific scoring of the full corpus; the biencoder shortlist reduces that second-stage work to ten candidates.

Retrieval results

Two target clauses were outside the baseline top three but moved to rank one after re-ranking. The scores below are the observed values in the saved notebook output.

Evidence query	PDF page	Rank B → R	Cosine	Cross score
Failed-attempt count after a new secret	33	4 → 1	0.5142	0.5854
Email confirmation/recovery-code exception	31	7 → 1	0.4775	2.9612

For the counter query, the baseline selected p036_c05, p035_c05, and p041_c03; the re-ranked selection was p033_c02, p042_c01, and p035_c05. For the email-code query, the respective selections were p059_c01, p059_c03, p059_c02 and p031_c03, p059_c03, p060_c01.

Generated answers

Retrieval improvement did not translate equally into answer quality. For the counter query, the baseline generated an unrelated rate-limiting fragment. With the recovered clause, the re-ranked answer was: "SHALL NOT reset the failed authentication count." For the email exception, the baseline answered "Yes" and the re-ranked answer was "no". The latter has the right

direction but omits the document's reason: confirmation and recovery codes are not authentication processes. The answer comparison therefore supports a retrieval improvement, not a claim that every generated answer became complete.

Latency

	Baseline		Re-ranked	
Query	**Generation**	**End-to-end**	**Generation**	**End-to-end**
Across the five retrieval tests, median baseline retrieval was 15.38 ms and median cross-encoder scoring of the top ten was 219.61 ms. Paired answer-run timings are shown separately below. End-to-end time includes the retrieval path and generation; model loading is excluded.				
Counter clause	6191.15 ms	6203.76 ms	1511.63 ms	1757.86 ms
Email exception	990.45 ms	1006.36 ms	848.39 ms	1189.13 ms

These are measurements from a small local run, not general performance estimates. The first generation call was much slower than the next, so warm-up is a likely influence. The data do not establish that re-ranking is universally faster.

Limitations and conclusion

The assignment requests at least two document-specific failure queries and gives keyword traps, exceptions, and negation as examples. The notebook contains two supported recoveries: an exception and a negation/constraint case. It does not establish a separate keyword-trap test, so that case remains to be tested. It also tests an AAL3 syncable-authenticator query whose target clause on page 20 was absent from the top ten. Re-ranking cannot recover evidence the bi-encoder did not retrieve. In this experiment, re-ranking was useful for refining a shortlist, but candidate coverage and answer completeness still need checking.

GitHub - https://github.com/Ashish-Beniwal004/Assignment1_LMA.git